

# Partial Fractions and Improper Integrals

ANSWER KEY



## Partial fraction decomposition: distinct linear factors

- 1 Decompose  $\frac{5x-1}{(x-1)(x+2)}$  into partial fractions.

$$\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}. \quad 5x-1 = A(x+2) + B(x-1). \quad \text{At } x=1: 4 = 3A \Rightarrow A = \frac{4}{3}. \quad \text{At } x=-2: -11 = -3B \Rightarrow B = \frac{11}{3}. \quad \text{So } \frac{4/3}{x-1} + \frac{11/3}{x+2}.$$

- 2 Evaluate  $\int \frac{3x+5}{x^2+x-2} dx$  using partial fractions (note  $x^2+x-2 = (x+2)(x-1)$ ).

$$\frac{3x+5}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}. \quad 3x+5 = A(x-1) + B(x+2). \quad \text{At } x=1: 8 = 3B \Rightarrow B = \frac{8}{3}. \quad \text{At } x=-2: -1 = -3A \Rightarrow A = \frac{1}{3}. \quad \int \left( \frac{1/3}{x+2} + \frac{8/3}{x-1} \right) dx = \frac{1}{3} \ln|x+2| + \frac{8}{3} \ln|x-1| + C.$$

- 3 Decompose  $\frac{x+7}{x^2-x-6}$  into partial fractions (note  $x^2-x-6 = (x-3)(x+2)$ ).

$$\frac{x+7}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}. \quad x+7 = A(x+2) + B(x-3). \quad \text{At } x=3: 10 = 5A \Rightarrow A = 2. \quad \text{At } x=-2: 5 = -5B \Rightarrow B = -1. \quad \text{So } \frac{2}{x-3} - \frac{1}{x+2}.$$

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## Partial fractions with a repeated linear factor

- 4 Decompose  $\frac{2x+3}{(x-1)^2}$  into partial fractions.

$$\frac{2x+3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}. \quad 2x+3 = A(x-1) + B. \quad \text{At } x=1: 5 = B. \quad \text{Comparing } x \text{ coefficients: } A = 2. \quad \text{So } \frac{2}{x-1} + \frac{5}{(x-1)^2}.$$

- 5 Evaluate  $\int \frac{2x+3}{(x-1)^2} dx$  using the decomposition found above.

$$\int \left( \frac{2}{x-1} + \frac{5}{(x-1)^2} \right) dx = 2 \ln|x-1| - \frac{5}{x-1} + C.$$

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## Partial fractions with an irreducible quadratic factor

- 6 Set up (but do not solve for constants) the partial fraction decomposition of  $\frac{3x^2+2}{(x-1)(x^2+4)}$ .

Since  $x^2+4$  is irreducible over the reals, the decomposition has the form  $\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$ .

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## Improper integrals: infinite limits

- 7 Evaluate  $\int_1^\infty \frac{1}{x^2} dx$ , or show it diverges.

$$\int_1^\infty \frac{1}{x^2} dx = \lim_{b \rightarrow \infty} \left[ -\frac{1}{x} \right]_1^b = \lim_{b \rightarrow \infty} \left( -\frac{1}{b} + 1 \right) = 0 + 1 = 1. \quad \text{Converges to 1.}$$

- 8 Evaluate  $\int_1^{\infty} \frac{1}{x} dx$ , or show it diverges.

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{b \rightarrow \infty} [\ln x]_1^b = \lim_{b \rightarrow \infty} (\ln b - 0) = \infty. \text{ Diverges.}$$

- 9 Evaluate  $\int_0^{\infty} e^{-2x} dx$ , or show it diverges.

$$\int_0^{\infty} e^{-2x} dx = \lim_{b \rightarrow \infty} \left[-\frac{1}{2}e^{-2x}\right]_0^b = \lim_{b \rightarrow \infty} \left(-\frac{1}{2}e^{-2b} + \frac{1}{2}\right) = 0 + \frac{1}{2} = \frac{1}{2}. \text{ Converges to } \frac{1}{2}.$$

- 10 Evaluate  $\int_{-\infty}^0 e^x dx$ , or show it diverges.

$$\int_{-\infty}^0 e^x dx = \lim_{a \rightarrow -\infty} [e^x]_a^0 = \lim_{a \rightarrow -\infty} (1 - e^a) = 1 - 0 = 1. \text{ Converges to } 1.$$

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### Improper integrals: discontinuous integrands

- 11 Evaluate  $\int_0^1 \frac{1}{\sqrt{x}} dx$ , or show it diverges (note the integrand is undefined at  $x = 0$ ).

$$\int_0^1 x^{-1/2} dx = \lim_{a \rightarrow 0^+} [2\sqrt{x}]_a^1 = \lim_{a \rightarrow 0^+} (2 - 2\sqrt{a}) = 2 - 0 = 2. \text{ Converges to } 2.$$

- 12 Evaluate  $\int_0^1 \frac{1}{x} dx$ , or show it diverges.

$$\int_0^1 \frac{1}{x} dx = \lim_{a \rightarrow 0^+} [\ln x]_a^1 = \lim_{a \rightarrow 0^+} (0 - \ln a) = \infty. \text{ Diverges.}$$

- 13 Evaluate  $\int_0^2 \frac{1}{(x-2)^2} dx$ , or show it diverges (the integrand is undefined at  $x = 2$ ).

$$\int_0^2 (x-2)^{-2} dx = \lim_{b \rightarrow 2^-} \left[-\frac{1}{x-2}\right]_0^b = \lim_{b \rightarrow 2^-} \left(-\frac{1}{b-2} - \frac{1}{2}\right) = \infty. \text{ Diverges.}$$

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### The p-test for improper integrals

- 14 State the p-test for  $\int_1^{\infty} \frac{1}{x^p} dx$ , and use it to determine convergence for  $p = 3$  and  $p = 0.5$ .

$$\int_1^{\infty} \frac{1}{x^p} dx \text{ converges if } p > 1 \text{ and diverges if } p \leq 1. \text{ For } p = 3 > 1: \text{ converges. For } p = 0.5 \leq 1: \text{ diverges.}$$

- 15 State the analogous p-test for  $\int_0^1 \frac{1}{x^p} dx$  (a discontinuity at  $x = 0$ ), and determine convergence for  $p = \frac{1}{2}$  and  $p = 2$ .

$$\int_0^1 \frac{1}{x^p} dx \text{ converges if } p < 1 \text{ and diverges if } p \geq 1 \text{ (opposite condition from the infinite-limit case). For } p = \frac{1}{2} < 1: \text{ converges. For } p = 2 \geq 1: \text{ diverges.}$$

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### Combining partial fractions with improper integrals

- 16 Evaluate  $\int_0^{\infty} \frac{1}{(x+1)(x+2)} dx$  using partial fractions.

$$\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}. \int_0^b \left(\frac{1}{x+1} - \frac{1}{x+2}\right) dx = \left[\ln \frac{x+1}{x+2}\right]_0^b = \ln \frac{b+1}{b+2} - \ln \frac{1}{2}. \text{ As } b \rightarrow \infty, \frac{b+1}{b+2} \rightarrow 1, \text{ so the limit is } \ln 1 - \ln \frac{1}{2} = \ln 2. \text{ Converges to } \ln 2.$$

### Three distinct linear factors

- 17 Decompose  $\frac{6x^2 - 3x - 1}{x(x-1)(x+1)}$  into partial fractions.

$\frac{6x^2 - 3x - 1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$ .  $6x^2 - 3x - 1 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$ . At  $x = 0$ :  $-1 = -A \Rightarrow A = 1$ . At  $x = 1$ :  $2 = 2B \Rightarrow B = 1$ . At  $x = -1$ :  $8 = 2C \Rightarrow C = 4$ . So  $\frac{1}{x} + \frac{1}{x-1} + \frac{4}{x+1}$ .

- 18 Evaluate  $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)} dx$  using the decomposition above.

$$\int \left( \frac{1}{x} + \frac{1}{x-1} + \frac{4}{x+1} \right) dx = \ln|x| + \ln|x-1| + 4\ln|x+1| + C.$$

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### Visualizing convergence: area under the curve

- 19 The convergent improper integral  $\int_1^{\infty} \frac{1}{x^2} dx = 1$  (shown earlier) means the infinite region under  $y = \frac{1}{x^2}$  for  $x \geq 1$  has FINITE area. Explain intuitively why this is possible even though the region extends forever.

Even though the region is infinitely long, the height  $\frac{1}{x^2}$  shrinks fast enough as  $x \rightarrow \infty$  that the accumulated area still converges to a finite total — the 'tail' contributes less and less area, summing to a bounded amount, similar to how a convergent geometric series sums to a finite value despite infinitely many terms.

- 20 By contrast,  $\int_1^{\infty} \frac{1}{x} dx$  diverges. Compare the graphs of  $y = \frac{1}{x}$  and  $y = \frac{1}{x^2}$  on  $[1, 10]$  and explain why  $\frac{1}{x}$  decays too slowly for the area to stay finite.

Both curves decrease, but  $\frac{1}{x}$  stays larger than  $\frac{1}{x^2}$  for all  $x > 1$  (e.g. at  $x = 10$ : 0.1 vs 0.01). This slower decay means the accumulated area under  $\frac{1}{x}$  keeps growing without bound as  $x \rightarrow \infty$ , while  $\frac{1}{x^2}$ 's faster decay allows its total area to converge.

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### Improper integrals with both an infinite bound and a discontinuity

- 21 Evaluate  $\int_0^{\infty} \frac{1}{\sqrt{x}(1+x)} dx$  is a classic doubly-improper integral (discontinuous at  $x = 0$  AND infinite upper bound). Explain how such an integral must be split to evaluate it.

Split at any convenient interior point, e.g.  $x = 1$ :  $\int_0^{\infty} = \int_0^1 + \int_1^{\infty}$ . Each piece is evaluated as its own limit (one with  $a \rightarrow 0^+$ , the other with  $b \rightarrow \infty$ ), and the original integral converges only if BOTH pieces converge.

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### Comparison test for improper integrals

- 22 Use a direct comparison to argue that  $\int_1^{\infty} \frac{1}{x^2+1} dx$  converges, without computing it exactly.

For  $x \geq 1$ :  $\frac{1}{x^2+1} < \frac{1}{x^2}$ . Since  $\int_1^{\infty} \frac{1}{x^2} dx$  converges (shown earlier,  $p = 2 > 1$ ), and the given integrand is smaller and positive,  $\int_1^{\infty} \frac{1}{x^2+1} dx$  also converges by comparison.

- 23 Use a direct comparison to argue that  $\int_1^{\infty} \frac{2+\sin x}{x} dx$  diverges.

Since  $-1 \leq \sin x \leq 1$ , we have  $2 + \sin x \geq 1$ , so  $\frac{2+\sin x}{x} \geq \frac{1}{x}$  for  $x \geq 1$ . Since  $\int_1^{\infty} \frac{1}{x} dx$  diverges, and the given integrand is at least as large,  $\int_1^{\infty} \frac{2+\sin x}{x} dx$  also diverges by comparison.

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### Multiple choice — easy 15

- 24 For  $\frac{7x+5}{(x+3)(x-4)} = \frac{A}{x+3} + \frac{B}{x-4}$ , find  $A$ .

- a.  $-\frac{16}{7}$
- b.  $\frac{23}{7}$
- c.  $\frac{16}{7}$  ✓
- d.  $\frac{33}{7}$

Using the cover-up method, set  $x = -3$ :  $A = \frac{7(-3)+5}{-3-(4)} = \frac{-16}{-7} = \frac{16}{7}$ .

- 25 For  $\frac{2x}{(x+3)(x-3)} = \frac{A}{x+3} + \frac{B}{x-3}$ , find  $A$ .

- a. 1 ✓
- b. -1
- c.  $\frac{3}{2}$
- d. 2

Using the cover-up method, set  $x = -3$ :  $A = \frac{2(-3)}{-3-(3)} = \frac{-6}{-6} = 1$ .

- 26 For  $\frac{7x-5}{(x)(x+5)} = \frac{A}{x} + \frac{B}{x+5}$ , find  $A$ .

- a. 8
- b. -1 ✓
- c. 1
- d. 0

Using the cover-up method, set  $x = 0$ :  $A = \frac{7(0)-5}{0-(-5)} = \frac{-5}{5} = -1$ .

27 For  $\frac{7x+2}{(x+6)(x+5)} = \frac{A}{x+6} + \frac{B}{x+5}$ , find A.

a. 40 ✓

b. -33

c. -40

d. 41

Using the cover-up method, set  $x = -6$ :  $A = \frac{7(-6)+2}{-6-(-5)} = \frac{-40}{-1} = 40$ .

28 For  $\frac{2x-6}{(x)(x+4)} = \frac{A}{x} + \frac{B}{x+4}$ , find A.

a.  $\frac{7}{2}$

b.  $-\frac{3}{2}$  ✓

c.  $\frac{3}{2}$

d.  $-\frac{1}{2}$

Using the cover-up method, set  $x = 0$ :  $A = \frac{2(0)-6}{0-(-4)} = \frac{-6}{4} = -\frac{3}{2}$ .

29 For  $\frac{6x-7}{(x+6)(x-4)} = \frac{A}{x+6} + \frac{B}{x-4}$ , find A.

a.  $\frac{53}{10}$

b.  $\frac{17}{10}$

c.  $\frac{43}{10}$  ✓

d.  $-\frac{43}{10}$

Using the cover-up method, set  $x = -6$ :  $A = \frac{6(-6)-7}{-6-(4)} = \frac{-43}{-10} = \frac{43}{10}$ .

30 For  $\frac{7x+1}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2}$ , find A.

a. 23

b. -22

c. -15

d. 22 ✓

Using the cover-up method, set  $x = 3$ :  $A = \frac{7(3)+1}{3-(2)} = \frac{22}{1} = 22$ .

31 For  $\frac{2x-8}{(x-2)(x)} = \frac{A}{x-2} + \frac{B}{x}$ , find  $A$ .

a. 2

b.  $-2$  ✓

c. 4

d.  $-1$

Using the cover-up method, set  $x = 2$ :  $A = \frac{2(2)-8}{2-(0)} = \frac{-4}{2} = -2$ .

32 For  $\frac{2x-7}{(x-1)(x+4)} = \frac{A}{x-1} + \frac{B}{x+4}$ , find  $A$ .

a.  $-1$  ✓

b. 1

c. 0

d. 3

Using the cover-up method, set  $x = 1$ :  $A = \frac{2(1)-7}{1-(-4)} = \frac{-5}{5} = -1$ .

33 For  $\frac{6x+9}{(x+1)(x-5)} = \frac{A}{x+1} + \frac{B}{x-5}$ , find  $A$ .

a.  $\frac{1}{2}$

b.  $\frac{13}{2}$

c.  $-\frac{1}{2}$  ✓

d.  $-1$

Using the cover-up method, set  $x = -1$ :  $A = \frac{6(-1)+9}{-1-(5)} = \frac{3}{-6} = -\frac{1}{2}$ .

34 For  $\frac{5x-4}{(x-3)(x+5)} = \frac{A}{x-3} + \frac{B}{x+5}$ , find A.

- a.  $-\frac{11}{8}$
- b.  $\frac{29}{8}$
- c.  $\frac{19}{8}$
- d.  $\frac{11}{8}$  ✓

Using the cover-up method, set  $x = 3$ :  $A = \frac{5(3)-4}{3-(-5)} = \frac{11}{8} = \frac{11}{8}$ .

35 For  $\frac{3x+3}{(x+3)(x-5)} = \frac{A}{x+3} + \frac{B}{x-5}$ , find A.

- a.  $\frac{7}{4}$
- b.  $-\frac{3}{4}$
- c.  $\frac{9}{4}$
- d.  $\frac{3}{4}$  ✓

Using the cover-up method, set  $x = -3$ :  $A = \frac{3(-3)+3}{-3-(5)} = \frac{-6}{-8} = \frac{3}{4}$ .

36 For  $\frac{4x}{(x)(x-3)} = \frac{A}{x} + \frac{B}{x-3}$ , find A.

- a.  $-\frac{1}{2}$
- b. 0 ✓
- c. 1
- d. 4

Using the cover-up method, set  $x = 0$ :  $A = \frac{4(0)}{0-(3)} = \frac{0}{-3} = 0$ .

37 For  $\frac{3x-4}{(x-4)(x-5)} = \frac{A}{x-4} + \frac{B}{x-5}$ , find A.

- a. -8 ✓
- b. 8
- c. -7
- d. 11

Using the cover-up method, set  $x = 4$ :  $A = \frac{3(4)-4}{4-(5)} = \frac{8}{-1} = -8$ .

38 For  $\frac{8x+9}{(x-6)(x-4)} = \frac{A}{x-6} + \frac{B}{x-4}$ , find A.

a.  $-\frac{57}{2}$

b.  $\frac{57}{2}$  ✓

c.  $-\frac{41}{2}$

d.  $\frac{59}{2}$

Using the cover-up method, set  $x = 6$ :  $A = \frac{8(6)+9}{6-(4)} = \frac{57}{2} = \frac{57}{2}$ .

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### Multiple choice — medium 15

39 Evaluate  $\int_1^{\infty} \frac{2}{x^3} dx$ , or show it diverges.

a.  $-1$

b.  $1$  ✓

c.  $2$

d.  $\frac{2}{3}$

$$\int_1^{\infty} 2x^{-3} dx = \lim_{b \rightarrow \infty} \left[ \frac{2x^{-2}}{-2} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-2}{(2)b^2} + \frac{2}{2} \right) = 0 + \frac{2}{2} = 1. \text{ Converges.}$$

40 Evaluate  $\int_1^{\infty} \frac{7}{x^2} dx$ , or show it diverges.

a.  $\frac{7}{2}$

b.  $7$  ✓

c.  $8$

d.  $-7$

$$\int_1^{\infty} 7x^{-2} dx = \lim_{b \rightarrow \infty} \left[ \frac{7x^{-1}}{-1} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-7}{(1)b^1} + \frac{7}{1} \right) = 0 + \frac{7}{1} = 7. \text{ Converges.}$$

41 Evaluate  $\int_1^{\infty} \frac{9}{x^6} dx$ , or show it diverges.

a.  $\frac{9}{5}$  ✓

b.  $\frac{3}{2}$

c.  $-\frac{9}{5}$

d.  $\frac{14}{5}$

$$\int_1^{\infty} 9x^{-6} dx = \lim_{b \rightarrow \infty} \left[ \frac{9x^{-5}}{-5} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-9}{(5)b^5} + \frac{9}{5} \right) = 0 + \frac{9}{5} = \frac{9}{5}. \text{ Converges.}$$

42 Evaluate  $\int_1^{\infty} \frac{8}{x^2} dx$ , or show it diverges.

a.  $-8$

b.  $4$

c.  $8$  ✓

d.  $9$

$$\int_1^{\infty} 8x^{-2} dx = \lim_{b \rightarrow \infty} \left[ \frac{8x^{-1}}{-1} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-8}{(1)b^1} + \frac{8}{1} \right) = 0 + \frac{8}{1} = 8. \text{ Converges.}$$

43 Evaluate  $\int_1^{\infty} \frac{3}{x^3} dx$ , or show it diverges.

a.  $-\frac{3}{2}$

b.  $\frac{3}{2}$  ✓

c.  $\frac{5}{2}$

d.  $1$

$$\int_1^{\infty} 3x^{-3} dx = \lim_{b \rightarrow \infty} \left[ \frac{3x^{-2}}{-2} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-3}{(2)b^2} + \frac{3}{2} \right) = 0 + \frac{3}{2} = \frac{3}{2}. \text{ Converges.}$$

44 Evaluate  $\int_1^{\infty} \frac{3}{x^2} dx$ , or show it diverges.

a.  $\frac{3}{2}$

b.  $-3$

c.  $4$

d.  $3$  ✓

$$\int_1^{\infty} 3x^{-2} dx = \lim_{b \rightarrow \infty} \left[ \frac{3x^{-1}}{-1} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-3}{(1)b^1} + \frac{3}{1} \right) = 0 + \frac{3}{1} = 3. \text{ Converges.}$$

45 Evaluate  $\int_1^{\infty} \frac{6}{x^5} dx$ , or show it diverges.

a.  $-\frac{3}{2}$

b.  $\frac{6}{5}$

c.  $\frac{5}{2}$

d.  $\frac{3}{2}$  ✓

$$\int_1^{\infty} 6x^{-5} dx = \lim_{b \rightarrow \infty} \left[ \frac{6x^{-4}}{-4} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-6}{(4)b^4} + \frac{6}{4} \right) = 0 + \frac{6}{4} = \frac{3}{2}. \text{ Converges.}$$

46 Evaluate  $\int_1^{\infty} \frac{4}{x^2} dx$ , or show it diverges.

a. 5

b. -4

c. 2

d. 4 ✓

$$\int_1^{\infty} 4x^{-2} dx = \lim_{b \rightarrow \infty} \left[ \frac{4x^{-1}}{-1} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-4}{(1)b^1} + \frac{4}{1} \right) = 0 + \frac{4}{1} = 4. \text{ Converges.}$$

47 Evaluate  $\int_1^{\infty} \frac{9}{x^3} dx$ , or show it diverges.

a.  $-\frac{9}{2}$

b.  $\frac{11}{2}$

c. 3

d.  $\frac{9}{2}$  ✓

$$\int_1^{\infty} 9x^{-3} dx = \lim_{b \rightarrow \infty} \left[ \frac{9x^{-2}}{-2} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-9}{(2)b^2} + \frac{9}{2} \right) = 0 + \frac{9}{2} = \frac{9}{2}. \text{ Converges.}$$

48 Evaluate  $\int_1^{\infty} \frac{7}{x^6} dx$ , or show it diverges.

a.  $\frac{7}{6}$

b.  $\frac{7}{5}$  ✓

c.  $\frac{12}{5}$

d.  $-\frac{7}{5}$

$$\int_1^{\infty} 7x^{-6} dx = \lim_{b \rightarrow \infty} \left[ \frac{7x^{-5}}{-5} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-7}{(5)b^5} + \frac{7}{5} \right) = 0 + \frac{7}{5} = \frac{7}{5}. \text{ Converges.}$$

49 Evaluate  $\int_1^{\infty} \frac{2}{x^4} dx$ , or show it diverges.

a.  $\frac{2}{3}$  ✓

b.  $\frac{5}{3}$

c.  $\frac{1}{2}$

d.  $-\frac{2}{3}$

$$\int_1^{\infty} 2x^{-4} dx = \lim_{b \rightarrow \infty} \left[ \frac{2x^{-3}}{-3} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-2}{(3)b^3} + \frac{2}{3} \right) = 0 + \frac{2}{3} = \frac{2}{3}. \text{ Converges.}$$

50 Evaluate  $\int_1^{\infty} \frac{1}{x^6} dx$ , or show it diverges.

a.  $-\frac{1}{5}$

b.  $\frac{1}{6}$

c.  $\frac{1}{5}$  ✓

d.  $\frac{6}{5}$

$$\int_1^{\infty} 1x^{-6} dx = \lim_{b \rightarrow \infty} \left[ \frac{1x^{-5}}{-5} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-1}{(5)b^5} + \frac{1}{5} \right) = 0 + \frac{1}{5} = \frac{1}{5}. \text{ Converges.}$$

51 Evaluate  $\int_1^{\infty} \frac{8}{x^4} dx$ , or show it diverges.

a.  $\frac{8}{3}$  ✓

b.  $-\frac{8}{3}$

c.  $\frac{11}{3}$

d. 2

$$\int_1^{\infty} 8x^{-4} dx = \lim_{b \rightarrow \infty} \left[ \frac{8x^{-3}}{-3} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-8}{(3)b^3} + \frac{8}{3} \right) = 0 + \frac{8}{3} = \frac{8}{3}. \text{ Converges.}$$

52 Evaluate  $\int_1^{\infty} \frac{6}{x^3} dx$ , or show it diverges.

a. 4

b. 3 ✓

c. 2

d. -3

$$\int_1^{\infty} 6x^{-3} dx = \lim_{b \rightarrow \infty} \left[ \frac{6x^{-2}}{-2} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-6}{(2)b^2} + \frac{6}{2} \right) = 0 + \frac{6}{2} = 3. \text{ Converges.}$$

53 Evaluate  $\int_1^{\infty} \frac{3}{x^4} dx$ , or show it diverges.

a.  $\frac{3}{4}$

b. 1 ✓

c. 2

d. -1

$$\int_1^{\infty} 3x^{-4} dx = \lim_{b \rightarrow \infty} \left[ \frac{3x^{-3}}{-3} \right]_1^b = \lim_{b \rightarrow \infty} \left( \frac{-3}{(3)b^3} + \frac{3}{3} \right) = 0 + \frac{3}{3} = 1. \text{ Converges.}$$

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### Multiple choice — hard 15

54 Evaluate  $\int_0^1 \frac{1}{x^{\frac{2}{3}}} dx$ , or show it diverges (the integrand is undefined at  $x = 0$ ).

a.  $\frac{8}{3}$

b.  $\frac{5}{2}$

c.  $-\frac{5}{3}$

d.  $\frac{5}{3}$  ✓

Since  $p = \frac{2}{3} < 1$ , this converges (by the  $p$ -test at a discontinuity).  $\int_0^1 x^{-\frac{2}{3}} dx = \lim_{a \rightarrow 0^+} \left[ \frac{x^{\frac{1}{3}}}{\frac{1}{3}} \right]_a^1 = \frac{1}{\frac{1}{3}} - 0 = \frac{5}{3}.$

55 Evaluate  $\int_0^1 \frac{2}{x^{\frac{3}{4}}} dx$ , or show it diverges (the integrand is undefined at  $x = 0$ ).

a.  $\frac{8}{3}$  ✓

b.  $\frac{11}{3}$

c. 8

d.  $-\frac{8}{3}$

Since  $p = \frac{3}{4} < 1$ , this converges (by the  $p$ -test at a discontinuity).  $\int_0^1 2x^{-\frac{3}{4}} dx = \lim_{a \rightarrow 0^+} \left[ \frac{2x^{\frac{1}{4}}}{\frac{1}{4}} \right]_a^1 = \frac{2}{\frac{1}{4}} - 0 = \frac{8}{1}$ .

56 Evaluate  $\int_0^1 \frac{1}{x^{\frac{3}{4}}} dx$ , or show it diverges (the integrand is undefined at  $x = 0$ ).

a. -4

b. 5

c. 4 ✓

d.  $\frac{4}{3}$

Since  $p = \frac{3}{4} < 1$ , this converges (by the  $p$ -test at a discontinuity).  $\int_0^1 x^{-\frac{3}{4}} dx = \lim_{a \rightarrow 0^+} \left[ \frac{x^{\frac{1}{4}}}{\frac{1}{4}} \right]_a^1 = \frac{1}{\frac{1}{4}} - 0 = 4$ .

57 Evaluate  $\int_0^1 \frac{2}{x^{\frac{3}{2}}} dx$ , or show it diverges (the integrand is undefined at  $x = 0$ ).

a.  $\frac{9}{2}$

b. -4

c. 4 ✓

d. 5

Since  $p = \frac{3}{2} > 1$ , this converges (by the  $p$ -test at a discontinuity).  $\int_0^1 2x^{-\frac{3}{2}} dx = \lim_{a \rightarrow 0^+} \left[ \frac{2x^{-\frac{1}{2}}}{-\frac{1}{2}} \right]_a^1 = \frac{2}{-\frac{1}{2}} - 0 = -4$ .

58 Evaluate  $\int_0^1 \frac{6}{x^{\frac{4}{5}}} dx$ , or show it diverges (the integrand is undefined at  $x = 0$ ).

a.  $\frac{15}{2}$  ✓

b. 30

c.  $-\frac{15}{2}$

d.  $\frac{17}{2}$

Since  $p = \frac{4}{5} < 1$ , this converges (by the  $p$ -test at a discontinuity).  $\int_0^1 6x^{-\frac{4}{5}} dx = \lim_{a \rightarrow 0^+} \left[ \frac{6x^{\frac{1}{5}}}{\frac{1}{5}} \right]_a^1 = \frac{6}{\frac{1}{5}} - 0 = \frac{30}{1}$ .

59 For  $\frac{5x-1}{(x-3)^2} = \frac{A}{x-3} + \frac{B}{(x-3)^2}$ , find  $B$ .

a. 16

b. 4

c. 14 ✓

d. -14

Multiplying both sides by  $(x-3)^2$ :  $5x-1 = A(x-3) + B$ . Setting  $x=3$ :  $B = 5(3) - 1 = 14$ .

60 For  $\frac{5x-9}{(x-4)^2} = \frac{A}{x-4} + \frac{B}{(x-4)^2}$ , find  $B$ .

a. 11 ✓

b. -4

c. 13

d. -11

Multiplying both sides by  $(x-4)^2$ :  $5x-9 = A(x-4) + B$ . Setting  $x=4$ :  $B = 5(4) - 9 = 11$ .

61 For  $\frac{4x+8}{(x-5)^2} = \frac{A}{x-5} + \frac{B}{(x-5)^2}$ , find  $B$ .

a. 28 ✓

b. 12

c. 30

d. -28

Multiplying both sides by  $(x-5)^2$ :  $4x+8 = A(x-5) + B$ . Setting  $x=5$ :  $B = 4(5) + 8 = 28$ .

62 For  $\frac{5x-3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$ , find  $B$ .

a. -2

b. 4

c. 6

d. 2 ✓

Multiplying both sides by  $(x-1)^2$ :  $5x-3 = A(x-1) + B$ . Setting  $x=1$ :  $B = 5(1) - 3 = 2$ .

63 For  $\frac{5x+1}{(x+3)^2} = \frac{A}{x+3} + \frac{B}{(x+3)^2}$ , find  $B$ .

a. 6

b. 14

c. -14 ✓

d. -12

Multiplying both sides by  $(x+3)^2$ :  $5x+1 = A(x+3) + B$ . Setting  $x=-3$ :  $B = 5(-3) + 1 = -14$ .

64 Use a direct comparison to determine whether  $\int_1^{\infty} \frac{4+\sin x}{x} dx$  converges or diverges.

a. Diverges only because  $a=4$  happens to be even.

b. Diverges, since  $\frac{4+\sin x}{x} \geq \frac{3}{x}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{3}{x} dx$  diverges since  $p=1 \leq 1$ . ✓

c. Converges, since  $\frac{4+\sin x}{x} \leq \frac{5}{x}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{5}{x} dx$  converges since  $p=1 > 1$ .

d. Cannot be determined by comparison, since  $\sin x$  oscillates.

Since  $-1 \leq \sin x \leq 1$ : Diverges, since  $\frac{4+\sin x}{x} \geq \frac{3}{x}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{3}{x} dx$  diverges since  $p=1 \leq 1$ .

65 Use a direct comparison to determine whether  $\int_1^{\infty} \frac{2+\sin x}{x^2} dx$  converges or diverges.

a. Cannot be determined by comparison, since  $\sin x$  oscillates.

b. Diverges, since  $\frac{2+\sin x}{x^2} \geq \frac{1}{x^2}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{1}{x^2} dx$  diverges since  $p=2 \leq 1$ .

c. Converges, since  $\frac{2+\sin x}{x^2} \leq \frac{3}{x^2}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{3}{x^2} dx$  converges since  $p=2 > 1$ . ✓

d. Converges only because  $a=2$  happens to be even.

Since  $-1 \leq \sin x \leq 1$ : Converges, since  $\frac{2+\sin x}{x^2} \leq \frac{3}{x^2}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{3}{x^2} dx$  converges since  $p=2 > 1$ .

66 Use a direct comparison to determine whether  $\int_1^{\infty} \frac{2+\sin x}{x^3} dx$  converges or diverges.

a. Cannot be determined by comparison, since  $\sin x$  oscillates.

b. Diverges, since  $\frac{2+\sin x}{x^3} \geq \frac{1}{x^3}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{1}{x^3} dx$  diverges since  $p = 3 \leq 1$ .

c. Converges, since  $\frac{2+\sin x}{x^3} \leq \frac{3}{x^3}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{3}{x^3} dx$  converges since  $p = 3 > 1$ . ✓

d. Converges only because  $a = 2$  happens to be even.

Since  $-1 \leq \sin x \leq 1$ : Converges, since  $\frac{2+\sin x}{x^3} \leq \frac{3}{x^3}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{3}{x^3} dx$  converges since  $p = 3 > 1$ .

67 Use a direct comparison to determine whether  $\int_1^{\infty} \frac{5+\sin x}{x} dx$  converges or diverges.

a. Diverges only because  $a = 5$  happens to be even.

b. Diverges, since  $\frac{5+\sin x}{x} \geq \frac{4}{x}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{4}{x} dx$  diverges since  $p = 1 \leq 1$ . ✓

c. Cannot be determined by comparison, since  $\sin x$  oscillates.

d. Converges, since  $\frac{5+\sin x}{x} \leq \frac{6}{x}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{6}{x} dx$  converges since  $p = 1 > 1$ .

Since  $-1 \leq \sin x \leq 1$ : Diverges, since  $\frac{5+\sin x}{x} \geq \frac{4}{x}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{4}{x} dx$  diverges since  $p = 1 \leq 1$ .

68 Use a direct comparison to determine whether  $\int_1^{\infty} \frac{5+\sin x}{x^2} dx$  converges or diverges.

a. Cannot be determined by comparison, since  $\sin x$  oscillates.

b. Converges only because  $a = 5$  happens to be even.

c. Converges, since  $\frac{5+\sin x}{x^2} \leq \frac{6}{x^2}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{6}{x^2} dx$  converges since  $p = 2 > 1$ . ✓

d. Diverges, since  $\frac{5+\sin x}{x^2} \geq \frac{4}{x^2}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{4}{x^2} dx$  diverges since  $p = 2 \leq 1$ .

Since  $-1 \leq \sin x \leq 1$ : Converges, since  $\frac{5+\sin x}{x^2} \leq \frac{6}{x^2}$  for  $x \geq 1$ , and  $\int_1^{\infty} \frac{6}{x^2} dx$  converges since  $p = 2 > 1$ .